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Department of Computer Science and Engineering

Project Synopsis

on

AI-Based Deepfake & Misinformation Detection System

Submitted in Partial Fulfillment of the Requirement

For the Degree of

Bachelor of Technology

in

Artificial Intelligence & Machine Learning

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LUCKNOW

(ODD SEM, 2026-27)

ABSTRACT

The rapid advancement of Artificial Intelligence and Generative AI has made it easier to create deepfake videos, manipulated images, and synthetic audio that can appear authentic. At the same time, genuine media can be misused by presenting it with incorrect dates, locations, events, or misleading captions. Therefore, detecting whether media is fake is not sufficient for reliable misinformation detection.

This project proposes an AI-powered multimodal verification system capable of analyzing images, videos, audio, and related news claims. The system will use computer vision and deep learning techniques for detecting manipulated or AI-generated media. Video analysis will involve frame extraction, face detection, and temporal analysis, while audio will be processed using suitable audio features and classification models.

An additional Context Verification Module will identify the available context of a media item and compare it with the associated claim. Relevant evidence will be retrieved from reliable sources to verify the date, location, event, and claim. The final system will classify content as Likely Authentic, Likely Manipulated, AI-Generated, Misleading Context, or Unverified, along with confidence scores and supporting evidence.

The proposed system aims to provide an explainable and practical platform for detecting both fake media and misleading use of genuine media.

TABLE OF CONTENTS

1. Introduction

2. Related Works / Literature Survey / Review

3. Problem Statement

4. Proposed Methodology

5. Conclusions

6. References

1. INTRODUCTION

The increasing availability of Artificial Intelligence and Generative AI has created new challenges in digital media verification. Deepfake technology can manipulate faces, voices, and videos, making artificial content difficult to distinguish from genuine media. Along with synthetic media, another major form of misinformation occurs when genuine images or videos are taken out of their original context and presented as evidence for a different event or claim.

Recent research has highlighted the importance of identifying the original context of visual content rather than relying only on image-forgery detection. Research on visual misinformation has investigated automated image contextualization and the prediction of original meta-context to support fact-checking.

Therefore, this project combines two related but distinct tasks: Deepfake/Fake Media Detection and Context-Based Misinformation Verification. The system will analyze the authenticity of the media first and then verify whether the associated claim matches the available evidence and original context.

The proposed system will provide users with an easy-to-use web interface where they can upload an image, video, audio file, or related news content and receive an AI-assisted verification report.

2. RELATED WORKS / LITERATURE SURVEY / REVIEW

2.1 Deepfake Detection

Deepfake detection has become an important research area because AI-generated and manipulated media can be used for misinformation and other malicious purposes. Modern approaches use deep learning and computer vision techniques to identify visual and temporal inconsistencies in manipulated content. Vision Transformer-based methods have also been investigated for deepfake detection because of their ability to capture fine-grained visual relationships.

2.2 Visual Misinformation and Original Context

Research on automated image contextualization has focused on identifying the original meta-context of visual misinformation. Such work is relevant to the proposed system because the project does not only ask whether an image or video is fake, but also whether genuine media is being used in the correct context.

2.3 Context and Veracity Verification

Research on context and veracity prediction has proposed sequential approaches in which a system first predicts the true context of an image and then evaluates the veracity of its caption. This supports the proposed Context Verification Module.

2.4 Evidence-Based Multimodal Fact-Checking

Recent multimodal fact-checking research has explored retrieval-enhanced verification, where relevant evidence is retrieved and accumulated to verify image-text claims. This supports the proposed use of evidence retrieval instead of relying only on an AI-generated prediction.

2.5 Research Gap

Existing systems often focus mainly on deepfake detection or misinformation verification as separate tasks. The proposed project aims to combine both capabilities in one practical application: **Media Authenticity + Context Verification + Evidence-Based Verdict.**

3. PROBLEM STATEMENT

The major problem addressed by this project is the difficulty of determining whether digital media is authentic and whether it is being presented with accurate information.

The system aims to address the following problems:

- Detection of AI-generated images and manipulated media.
- Detection of deepfake videos and suspicious video frames.
- Detection of synthetic or AI-generated audio.
- Identification of manipulated facial content.
- Detection of old media being presented as a recent event.
- Identification of mismatches between media and its claimed event or location.
- Verification of claims associated with images and videos.
- Retrieval and presentation of supporting evidence.
- Providing an understandable explanation instead of only a Real/Fake label.

Main Objective

To develop an AI-powered multimodal system that can detect manipulated or AI-generated media and verify whether genuine media is being presented with the correct context and supporting evidence.

4. PROPOSED METHODOLOGY

The proposed system will be implemented as a web-based AI application. The user will upload an image, video, audio file, or related claim through a React-based interface. The file will be sent to a Python FastAPI backend for preprocessing and AI analysis.

4.1 Overall Architecture

User → React Web Application → Image / Video / Audio / News Claim → FastAPI Backend → Preprocessing → Image / Video / Audio Analysis → Deepfake Detection → Context Verification → Evidence Retrieval → Final Verification → Report + History

4.2 Media Input

The user will be able to upload images such as JPG/PNG, videos such as MP4/MOV, audio such as MP3/WAV, and text or captions associated with the media.

4.3 Image Analysis

Image → Preprocessing → Face Detection → Feature Extraction → CNN / ViT Model → Real / Manipulated / AI-Generated.

The system will generate a confidence score and, where possible, highlight suspicious regions using explainable AI techniques.

4.4 Video Analysis

Video → Frame Extraction → Face Detection → Face/Frame Analysis → CNN → Temporal Analysis → LSTM / Transformer → Overall Video Prediction.

Instead of analyzing the entire video as a single input, selected frames will be extracted and analyzed. Temporal analysis can then identify inconsistencies across frames.

4.5 Audio Analysis

Audio → Preprocessing → MFCC / Spectrogram → Feature Extraction → Audio Classification Model → Real / Synthetic Voice.

For videos containing audio, visual and audio predictions can be combined to produce a multimodal result.

4.6 Context Verification Module

This will be the main enhancement of the project. If a user uploads an authentic video with a claim such as “This video shows a recent incident”, the system will attempt to verify the event, location, date, and available evidence. Relevant evidence will be retrieved and compared with the current claim.

4.7 Final Verdict

Likely Authentic & Supported: Media appears genuine and available evidence supports the claim.

Likely Manipulated: Strong indicators of manipulation/deepfake are detected.

Likely AI-Generated: Strong indicators of synthetic generation are detected.

Misleading Context: Media may be genuine, but the associated date, location, event, or claim does not match available evidence.
Unverified: Insufficient reliable evidence is available.

4.8 Explainable AI

The system will provide confidence scores, suspicious video frames, suspicious image regions, detection reasons, and supporting evidence. Techniques such as Grad-CAM or saliency visualization can be considered for explaining vision-model predictions.

4.9 Database and History

PostgreSQL will store detection history including filename, media type, prediction, confidence, context result, evidence/source information, analysis timestamp, and report information.

4.10 Technology Stack

Layer	Technology
Frontend	React.js, Tailwind CSS, Chart.js
Backend	Python, FastAPI
AI/ML	PyTorch/TensorFlow, CNN, LSTM/Transformer, Vision Transformer
Computer Vision	OpenCV, Face Detection, Video Frame Extraction
Audio	Librosa, MFCC, Spectrogram Analysis
Database	PostgreSQL
Development	Git, GitHub
Deployment	Vercel/Netlify, Render/Railway

4.11 Model Development and Evaluation

Publicly available deepfake datasets such as UADFV (Uncompressed Facial Video Dataset), Celeb-DF, and DeeperForensics-1.0 can be considered for model development and evaluation. A practical implementation can start with a pretrained model and fine-tune it rather than building every model from scratch.

The model will be evaluated using standard metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

4.12 Development Phases

Phase 1: React website, upload page, result page, and history page.

Phase 2: Working image deepfake detection prototype.

Phase 3: Video frame extraction, face analysis, and prediction.

Phase 4: Audio processing and synthetic voice detection.

Phase 5: Context/news verification and evidence retrieval.

Phase 6: Explainable AI with heatmaps and suspicious frames.

Phase 7: PostgreSQL detection history.

Phase 8: Full React + FastAPI + AI integration.

Phase 9: Testing, documentation, and deployment.

5. CONCLUSIONS

The proposed project aims to develop a unified AI-based platform for deepfake detection and misinformation verification. Unlike conventional systems that only classify media as real or fake, the proposed system will also examine the context associated with genuine media.

The system will analyze images, videos, and audio using AI/ML and computer vision techniques and will further verify associated claims using context analysis and evidence retrieval. The expected outcome is a practical web application that provides users with a verification result, confidence score, explanation, and available supporting evidence.

The project therefore addresses two complementary questions: (1) Is the media real, manipulated, or AI-generated? and (2) If it is genuine, is it being presented with the correct information and context?

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